SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Lakshmipriya OS (192511) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

O.S Lakshmipriya

Signature of the student:

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
    a. Show how each attack leads to specific threats and risks
    b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
    a. Show differences and similarities
        i. Map how each affects the Data structure and Security level
        ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
    a. Show relationships between these concepts
    b. Analyze how each contributes to secure communication
    c. Include one real-world application for each

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Security attacks, threats, Risks & Security Goals.

Security attacks.
- Eavesdropping
- tampering
- Impersonation
- Malware
- Denial of service(DOS)

→

Threats
- Data interception
- Data modification
- Unauthorised access
- System disruption

→

Risks
- Loss of confidentiality
- Loss of integrity
- Loss of availability
- Financial

→

System goals
- confidentiality
- Integrity
- Availability

CNS - Assignment.

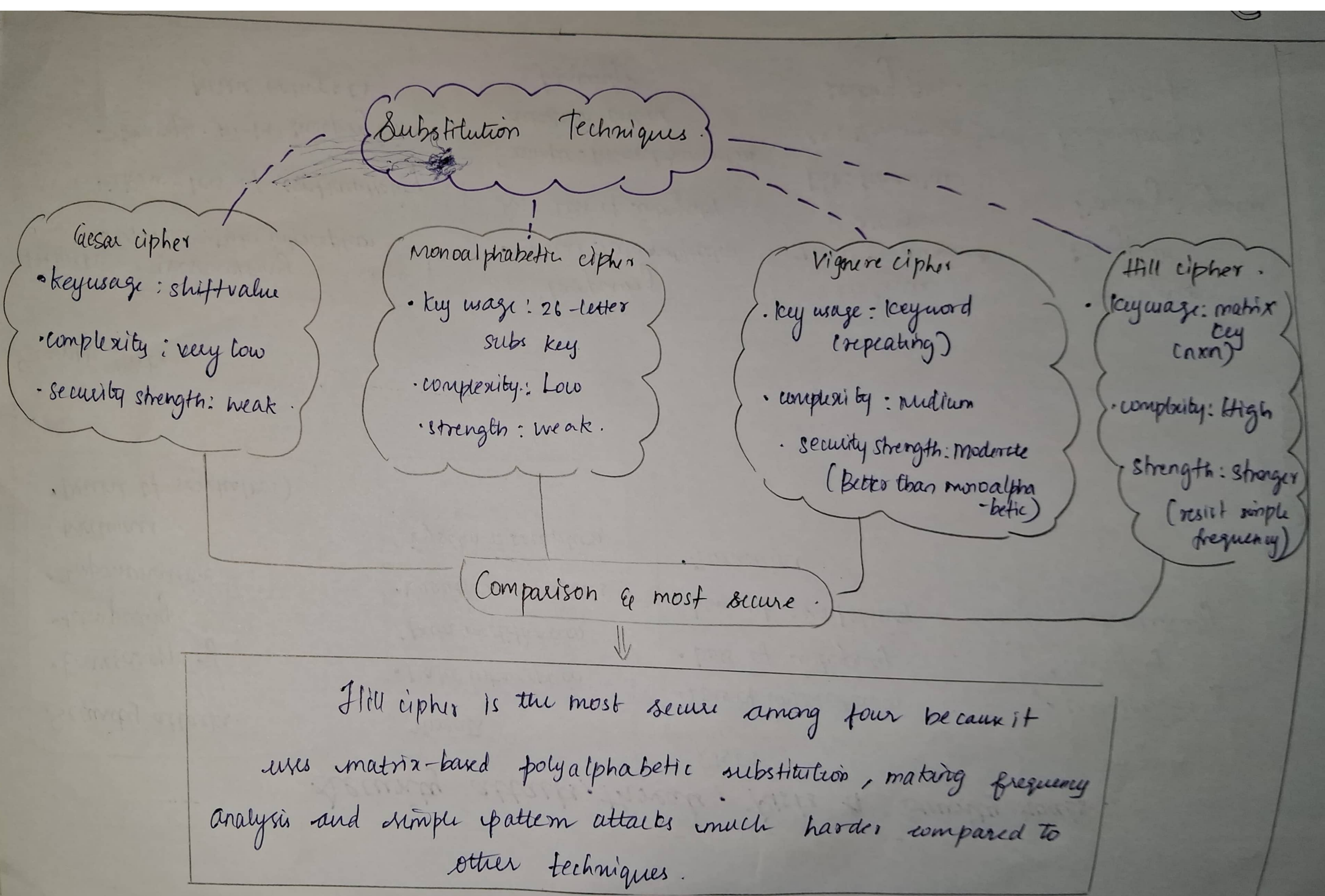
Example links.

Attacks: Eavesdropping
→ Threat: Data interception
→ Risk: Loss of confidentiality
→ Example: Wi-Fi snooping on public networks

Tampering
→ Threat: Data modification
→ Risk: Loss of integrity
→ Example: Altered transaction details in online banking.

Malware
Threat: unauthorised access
Risk: Financial
ex: Ransomware Locking files.

Mitigate
Encryption
Hashing & signature
Redundancy & backups.


Substitution Techniques
Caesar cipher
• key usage : shift value
• complexity : very low
• security strength: weak.
Monoalphabetic cipher
• key usage : 26-letter subs key
• complexity: Low
• strength : weak.
Vigenere cipher
• key usage : keyword (repeating)
• complexity : medium
• security strength: Moderate (Better than monoalpha-betic)
Hill cipher.
• key usage: matrix key (nxn)
• complexity: High
• strength: stronger (resist simple frequency)
Comparison & most secure.
Hill cipher is the most secure among four because it uses matrix-based polyalphabetic substitution, making frequency analysis and simple pattern attacks much harder compared to other techniques.

Classical Encryption schemes

Substitution technique

(eg. caesar, monoalphabetic, Vigenere, Hill)

Both are classical ciphers.

Transposition techniques

eg: Railfence, columnar, Playfair [combined])

Effects

- Datastructure : changes the actual characters (same length)
- Security level : Low to medium

- Data structure : Rearranges the order of characters
- Security level : Medium.

Similarities

- Both are classical.
- Both are Riversible
- Both can be combined for stronger security.

Comparison

- substitution changes the character value.
- Transposition changes the character position.

- substitution techniques are more vulnerable.
- Transposition is relatively stronger but can still be broken with pattern.

Steganography

Data hiding
Hides secret data inside another medium (image, audio, text).

uses

Watermarking
Embeds a visible/invisible mark to prove ownership.

enhances

Survivability
Ability of hidden data/mark to remain intact dispite attacks.

How they contribute to secure communication

- Conceals the existence of communication
- prevents detection.

- provides Integrity.
- helps in copyright protection.

- Ensures the hidden data/mark survives common attacks
- Maintain reliability of communication.

Real world applications

- Secret messages in Images.
  eg: image steganography

- Digital watermarking in videos, images, documents

- copyright protection for digital media
  eg: Netflix, youtube